

1.1.1

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Ecuación parabólica:

$$y(t) = y_0 + v_0 \sin \theta t - \frac{1}{2} g t^2 \quad (1)$$

$$y_0 = 0$$

$$x = v_x t$$

$$x = v_0 \cos \theta t \rightarrow t = \frac{x}{v_0 \cos \theta} \quad (2)$$

(2) en (1):

$$h = x \tan \theta - \frac{g x^2}{2 v_0^2 \cos^2 \theta}$$

Despejar v_0 :

$$x \tan \theta - h = \frac{g x^2}{2 v_0^2 \cos^2 \theta}$$

$$\Leftrightarrow 2 v_0^2 \cos^2 \theta = \frac{g x^2}{x \tan \theta - h}$$

$$v_0 = \sqrt{\frac{g x^2}{2(x \tan \theta - h) \cos^2 \theta}} \quad (3)$$

Ahora toca $N'_0(\theta) = 0$.

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$$N_0 = x \sqrt{\frac{g}{2}} \left[\underbrace{(x \tan \theta - h) \cos^2 \theta}_{f(x)} \right]^{-1/2}$$

$$N'_0 = \frac{-x}{2} \sqrt{\frac{g}{2}} \cdot f(x) \cdot f'(x) = 0$$

De (3) $f(x) \neq 0$.

Entonces

$$f'(x) = 0$$

$$\Leftrightarrow (x \sec^2 \theta)(\cos^2 \theta) + (x \tan \theta - h)(2 \cos \theta)(-\sin \theta) = 0$$

$$\Leftrightarrow x - 2(x \tan \theta - h) \cos \theta \sin \theta = 0$$

$$\Leftrightarrow x = 2x \tan \theta \cos \theta \sin \theta - 2h \cos \theta \sin \theta$$

$$\Leftrightarrow x = 2x \sin^2 \theta - h \sin(2\theta)$$

$$\Leftrightarrow x = x(1 - \cos(2\theta)) - h \sin(2\theta)$$

$$\Leftrightarrow x = x - x \cos(2\theta) - h \sin(2\theta)$$

$$\Leftrightarrow x \cos(2\theta) = -h \sin(2\theta)$$

$$\Leftrightarrow \frac{x}{h} = -\tan(2\theta) \quad (4)$$

$$\Leftrightarrow \frac{x}{h} = \tan(\pi - 2\theta) \rightarrow \boxed{\theta_{\min} = \frac{\pi}{2} - \frac{1}{2} \arctan\left(\frac{x}{h}\right)}$$

[1.2.]

Tenemos

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$$n_0 = \sqrt{\frac{gx^2}{(2l \tan \theta - h) \cos^2 \theta}} \quad (3)$$

$$\theta_{\min} = \frac{\pi}{2} - \frac{1}{2} \arctan(x/h) \quad (4)$$

No vamos a meter (4) en (3) porque es una pelada. Queda un bicho bien horrible. Así que vamos a sacar los prohibidos.

$$\text{Sea } \arctan(x/h) = \phi \quad (5)$$

Entonces (5) en (4):

$$\theta_{\min} = \frac{\pi - \phi}{2}$$

Queremos expresiones para $\cos(\theta_{\min})$ y $\tan(\theta_{\min})$ que no tengan la $\arctan$

$$\begin{aligned} \cos(\theta_{\min}) &= \cos\left(\frac{\pi}{2} - \frac{\phi}{2}\right) \\ &= \cancel{\cos(\pi/2)} \cos(\phi/2) + \cancel{\sin(\pi/2)} \sin(\phi/2) \\ &= \sin(\phi/2) \\ &= \sqrt{\frac{1 - \cos \phi}{2}} \quad (6) \end{aligned}$$

$$\sin(\theta_{\min}) = \sin\left(\frac{\pi}{2} - \frac{\phi}{2}\right)$$

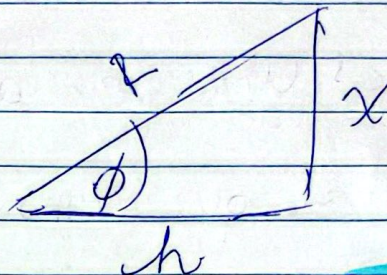
$$= \sin\left(\frac{\pi}{2}\right)\cos\left(\frac{\phi}{2}\right) - \sin\left(\frac{\phi}{2}\right)\cos\left(\frac{\pi}{2}\right)$$

$$= \cos\left(\frac{\phi}{2}\right)$$

$$= \sqrt{\frac{1 + \cos \phi}{2}} \quad (7)$$

De (5):

$$\tan \phi = \frac{x}{h}$$



$$R = (h^2 + x^2)^{1/2}$$

$$\cos \phi = \frac{h}{R} \quad (8)$$

$$\sin \phi = \frac{x}{R}$$

(8) en (6) y (7)

$$\cos(\theta_{\min}) = \sqrt{\frac{1 - \frac{h}{R}}{2}} \quad (9)$$

$$\sin(\theta_{\min}) = \sqrt{\frac{1 + \frac{h}{R}}{2}} \quad (10)$$

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(9) y (10) en el denominador (sin raíz)
de (3):

$$\Leftrightarrow \left(1 - \frac{h}{R}\right) \left(x \left[\frac{1 + h/R}{1 - h/R} \right]^{1/2} - h \right)$$

$$\Leftrightarrow \left(\frac{R-h}{R} \right) \left(\frac{x(R+h)^{1/2} - h(R-h)^{1/2}}{(R-h)^{1/2}} \right)$$

$$\Leftrightarrow \frac{1}{R} (R-h)^{1/2} \left(x(R+h)^{1/2} - h(R-h)^{1/2} \right)$$

$$\Leftrightarrow \frac{1}{R} \left\{ x[(R-h)(R+h)]^{1/2} - h(R-h) \right\}$$

$$\Leftrightarrow \frac{1}{R} \left\{ x \underbrace{(R^2 - h^2)}_{x^2}^{1/2} - Rh + h^2 \right\}$$

$$\Leftrightarrow \frac{1}{R} \left\{ x^2 - Rh + h^2 \right\}$$

$$\Leftrightarrow \frac{1}{R} \left\{ R^2 - Rh \right\}$$

$$\Leftrightarrow R - h \quad (11)$$

(11) en (3)

$$v_0 = \sqrt{\frac{gx^2}{R-h}}$$

$$\Rightarrow v_0 = \sqrt{\frac{gx^2(R+h)}{(R-h)(R+h)}}$$

$$= \frac{gx^2(R+h)}{R^2 - h^2}$$

$$= \sqrt{\frac{gx^2(R+h)}{x^2}}$$

$$= \sqrt{g(R+h)}$$

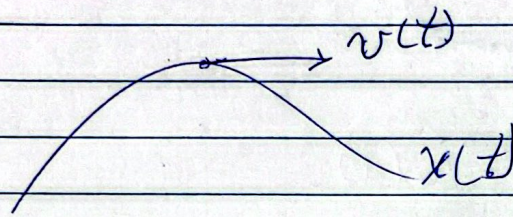
$$v_0 = \sqrt{g(\sqrt{x^2+h^2} + h)}$$

Problema #2

$$\vec{v}(t) = \hat{i} + 4t\hat{j} - 3t^2\hat{k}$$

$$t = 1.$$

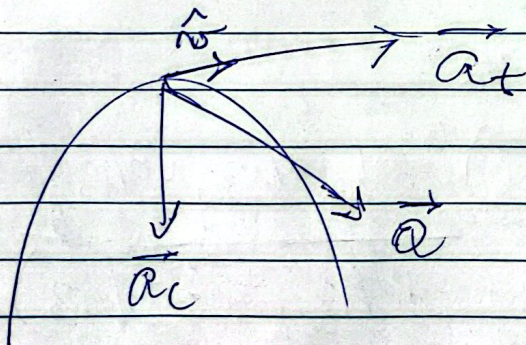
- ① El vector unitario en la dirección de la tangente de la trayectoria



$$\hat{v}(1) = \frac{\vec{v}(1)}{|\vec{v}(1)|} = \frac{(1, 4, -3)}{(1^2 + 4^2 + 3^2)^{1/2}}$$

$$\hat{v}(1) = \frac{1}{\sqrt{26}} (1, 4, -3)$$

- ② El vector aceleración tangencial



$$\vec{a} = \frac{d\vec{v}}{dt} = (1, 4t, -3t^2)'$$

$$= (0, 4, -6t)$$

$$\vec{a}(1) = (0, 4, -6)$$

$$\vec{a}(1) \cdot \hat{v}(1) = a_t$$

$$a_t = (0, 4, -6) \cdot (1, 4, -3) \frac{1}{\sqrt{26}}$$

$$= (0 + 16 + 18) \frac{1}{\sqrt{26}}$$

$$a_t = \frac{17\sqrt{26}}{13}$$

$$\vec{a}_t = a_t \cdot \hat{v}(1)$$

$$= \frac{17\sqrt{26}}{13} (1, 4, -3) \cdot \frac{1}{\sqrt{26}}$$

$$\vec{a}_t = \left(\frac{17}{13}, \frac{68}{13}, \frac{-51}{13} \right)$$

③ El vector aceleración normal

$$\vec{a} = \vec{a}_t + \vec{a}_n$$

$$\vec{a}_n = \vec{a} - \vec{a}_t$$

$$= (0, 4, -6) - \left(\frac{17}{13}, \frac{68}{13}, \frac{-51}{13} \right)$$

$$\vec{a}_n = \left(-\frac{17}{13}, -\frac{16}{13}, -\frac{27}{13} \right)$$

- ④ El vector unitario en la dirección de la normal de la trayectoria

$$\hat{a}_c = \frac{\vec{a}_c}{|\vec{a}_c|} = \frac{(-17/13, -16/13, -27/13)}{\left[\left(\frac{17}{13}\right)^2 + \left(\frac{16}{13}\right)^2 + \left(\frac{27}{13}\right)^2 \right]^{1/2}}$$

$$\boxed{\hat{a}_c = \frac{1}{7\sqrt{26}} (-17, -16, -27)}$$

- ⑤ El radio de curvatura ρ .

$$a_c = \frac{v^2}{\rho}$$

$$\rho = \frac{v^2}{a_c} = \frac{(\vec{v}(1) \cdot \vec{v}(1))}{[\vec{a}_c(1) \cdot \vec{a}_c(1)]^{1/2}}$$

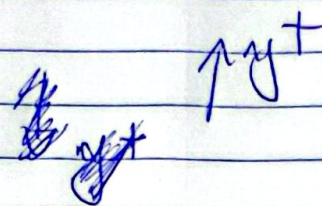
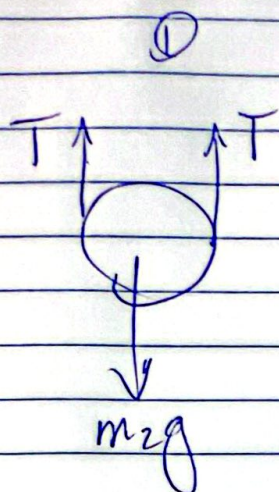
$$= \frac{1^2 + 4^2 + 3^2}{\left[\left(\frac{17}{13}\right)^2 + \left(\frac{16}{13}\right)^2 + \left(\frac{27}{13}\right)^2 \right]^{1/2}}$$

$$\left[\left(\frac{17}{13}\right)^2 + \left(\frac{16}{13}\right)^2 + \left(\frac{27}{13}\right)^2 \right]^{1/2}$$

$$\boxed{\rho = \frac{13\sqrt{26}}{7}}$$

Problema #3

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~~1m1g~~

$$① \quad 2T - m_2g = m_2a_2$$

$$② \quad T - m_1g = m_1a_1$$

$$-2f_2 + f_1:$$

$$2T - m_2g = m_2a_2$$

$$-2T + 2m_1g = -2m_1a_1$$

$$\Rightarrow 0 + 2m_1g - m_2g = m_2a_2 - 2m_1a_1$$

Per $a_1 = -2a_2$ Entorses

$$2m_1g - m_2g = m_2a_2 + 4m_1a_2$$

$$a_2 = \frac{2m_1g - m_2g}{4m_1 + m_2}$$

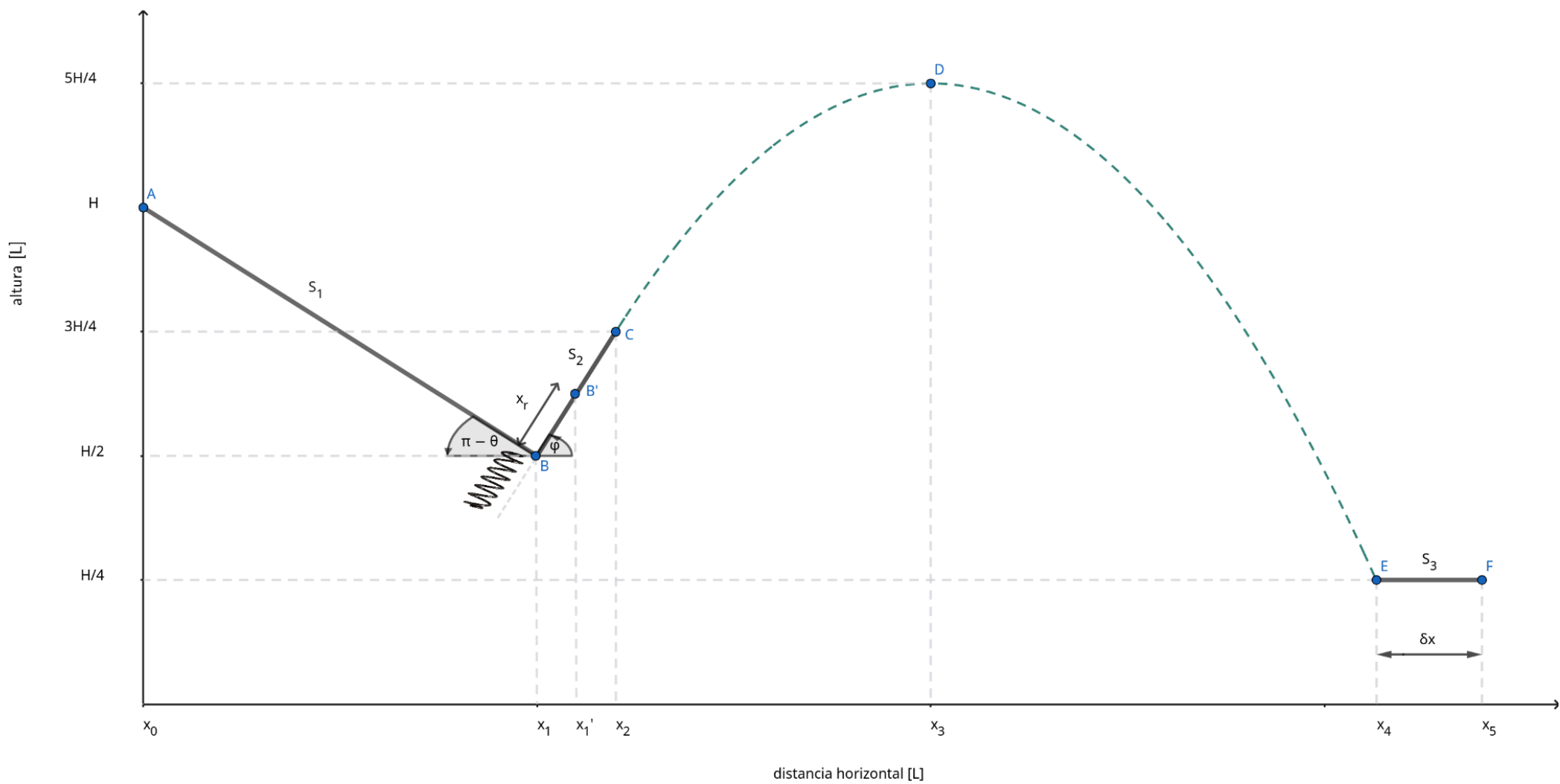
$$\rightarrow a_1 = -2a_2 = \left(\frac{2m_2g - 4m_1g}{4m_1 + m_2} \right)$$

($a_1 = 0$)
• $a_2 = 0$ si $2m_1 = m_2$

• $a_2 > 0$ ($a_1 < 0$)
si $2m_1 > m_2$

• $a_2 < 0$ ($a_1 > 0$)
si $2m_1 < m_2$

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[4.2]

Problema #4.

$$E_A = E_F + W_{fr}$$

$$E_A - W_{fr} = E_F$$

$$U_g^A + U_r^A + T^A - \vec{F}_N \cdot d = U_g^F$$

$$mgH + \frac{1}{2}Kx_r^2 + 0 - \mu mg \delta x = mg \frac{H}{4}$$

$$\frac{3}{4}mgH + \frac{1}{2}Kx_r^2 = \mu mg \delta x$$

$$x_r^2 = \frac{2}{K}(\mu mg \delta x - 3mgH/4)$$

$$x_r = \sqrt{\frac{2mg}{K} \left(\mu \delta x - \frac{3H}{4} \right)}$$

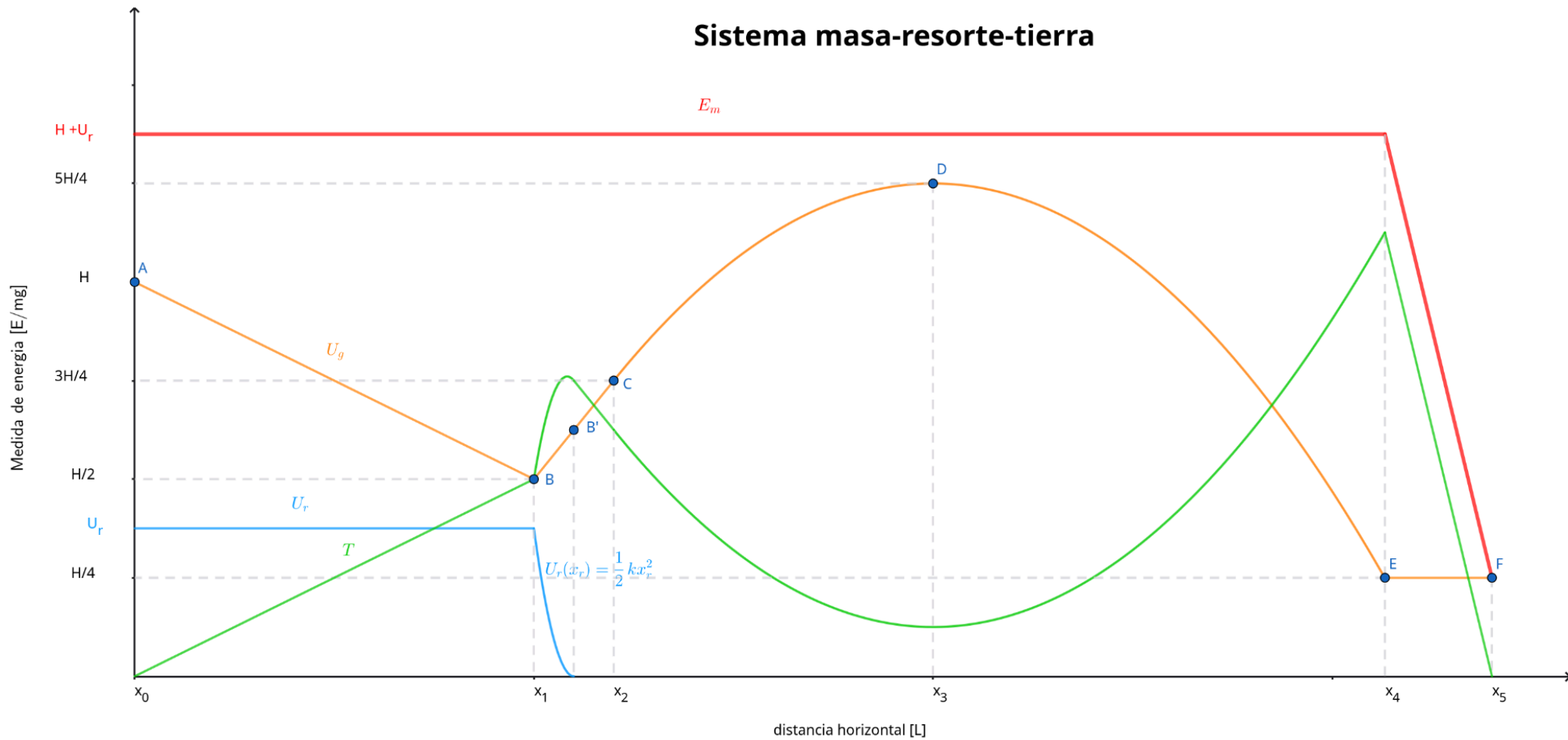
o bien

$$\delta x(x_r) = \frac{3H}{4\mu} + \frac{Kx_r^2}{2\mu mg}$$

$\delta x(x_r)$ es mínimo si $x_r = 0$.

O sea, si no hay resorte.

(O si está en equilibrio)



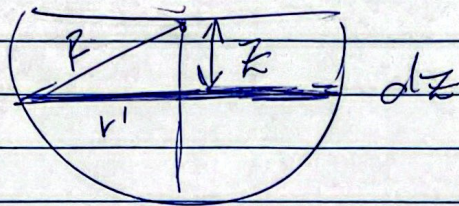
Nota: Lo único que se puede establecer con respecto a U_r es que debe ser $\geq H/4$ para que la partícula llegue a $5H/4$ en x_3 . Por lo que, aparte de cumplir dicha condición, su elección aquí es arbitraria, pues no sabemos nada de x_r ni de la constante k .

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Problema #5

$$W = \vec{F} \cdot d\vec{l}$$

Consideremos discos de grosor dz y radio r' como se muestra en la fig.



$$R^2 = z^2 + r'^2$$

$$r'^2 = R^2 - z^2$$

$$W = \vec{P} \cdot \vec{z}$$

$$dW = dP \cdot z$$

$$= dm \cdot g \cdot z$$

$$= dV \cdot g \cdot z$$

$$\rho = \frac{dm}{dV} = 1$$

$$\Leftrightarrow dm = dV$$

El volumen de cada disco es

$$A \cdot h = \pi r'^2 \cdot dz$$

$$= \pi (R^2 - z^2) dz$$

Entonces

$$dW = \pi (R^2 - z^2) dz \cdot g \cdot z$$

$$W = g\pi \int_0^R z(R^2 - z^2) dz$$

$$W = g\pi \left(\frac{R^2 z^2}{2} - \frac{z^4}{4} \right) \Big|_0^R =$$

$$\boxed{\frac{g\pi R^4}{4}}$$

(b) potencia

$$P = \frac{W}{t}$$
$$\boxed{P = \frac{g\pi R^4}{4t}}$$